



Lab #01

Github Identity, Student Portal Registration, and Ubuntu Server Installation

Name: Tayyaba Noor

Registration No: FA24B1-SE-068

Course Title: Cloud Computing

Course Code: BSE-410

Section: B

Session: 2024-2028

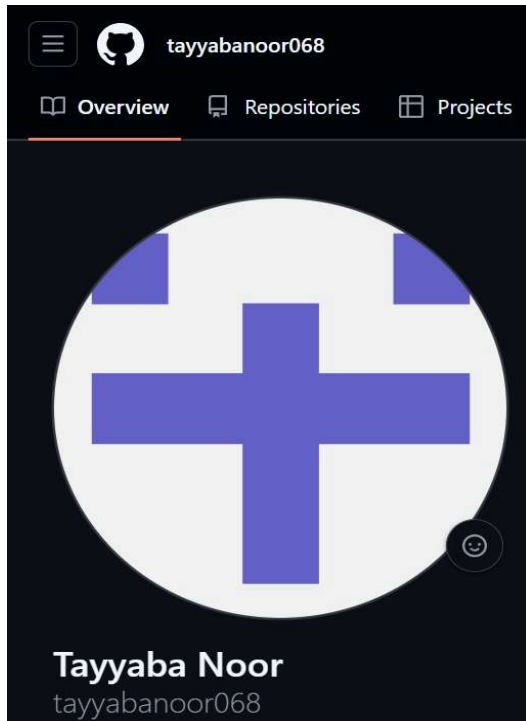
Github Username: tayyabanoor068

Fatima Jinnah Women University
Rawalpindi

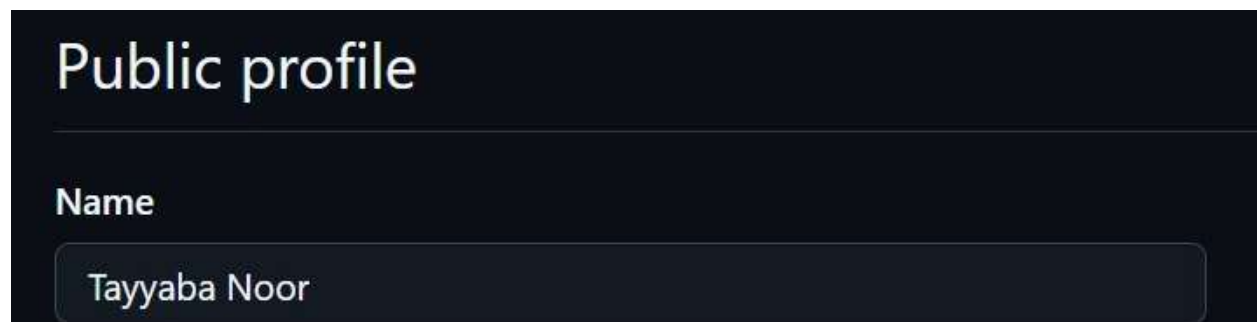
Task 1:

Create and configure your Github Account

- GitHub profile showing username tayyabanoor068



- GitHub settings showing complete actual name configured



Task 2:

Register in the Student Marks Portal

- Student Marks Portal initial login page.



Sign in to **GitHub**
to continue to **Firebase GitHub Login**

Username or email address

Password [Forgot password?](#)

Sign in

- GitHub profile details displayed inside the portal.

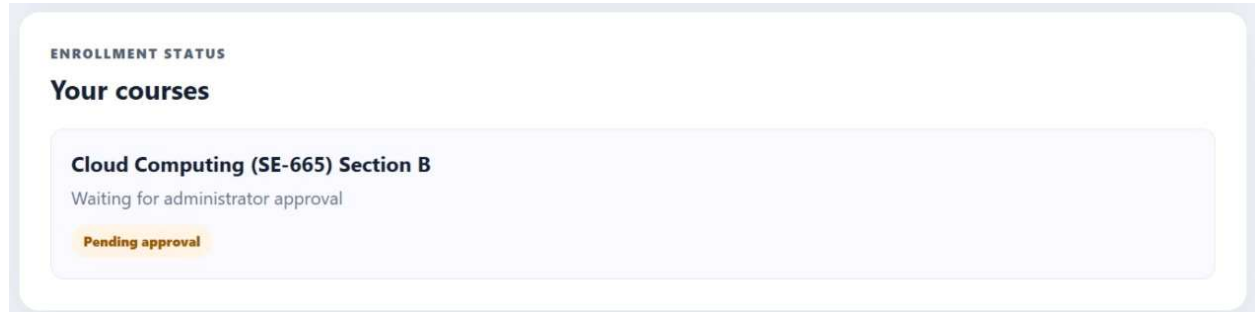
Course enrollments [Sign out](#)



Tayyaba Noor
2024-BSE-68

[+ Enroll in another course](#)

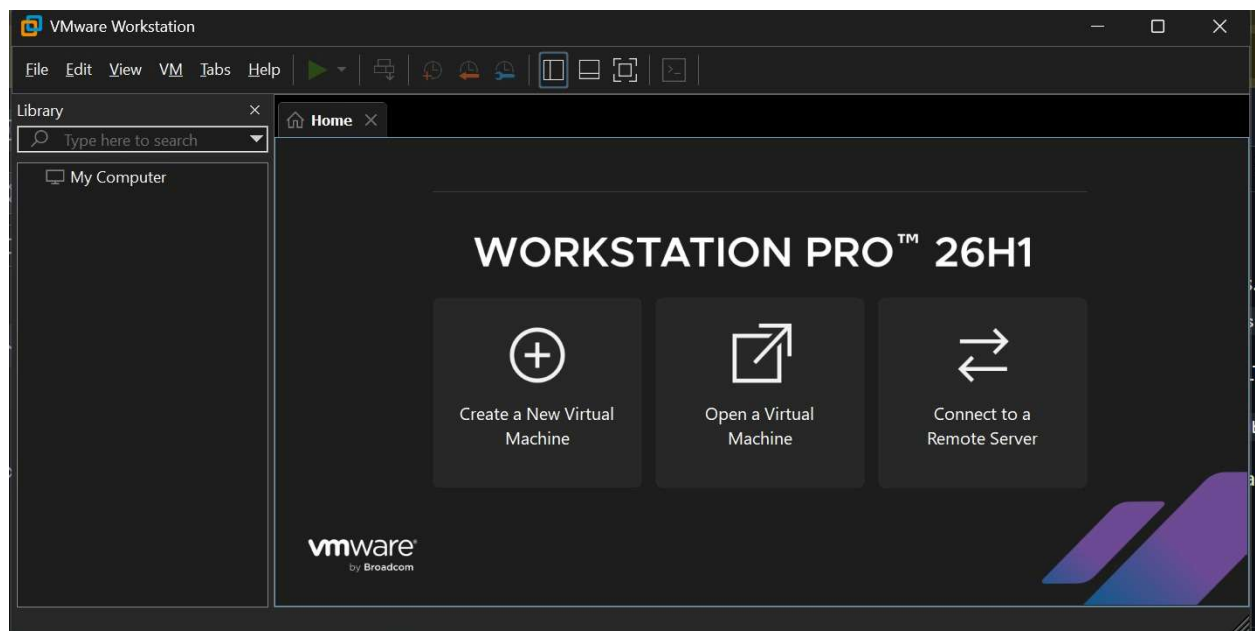
- Portal enrollment submission confirmation.



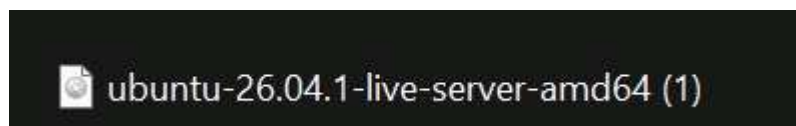
Task 3:

Download VMware Workstation Pro and Ubuntu Server

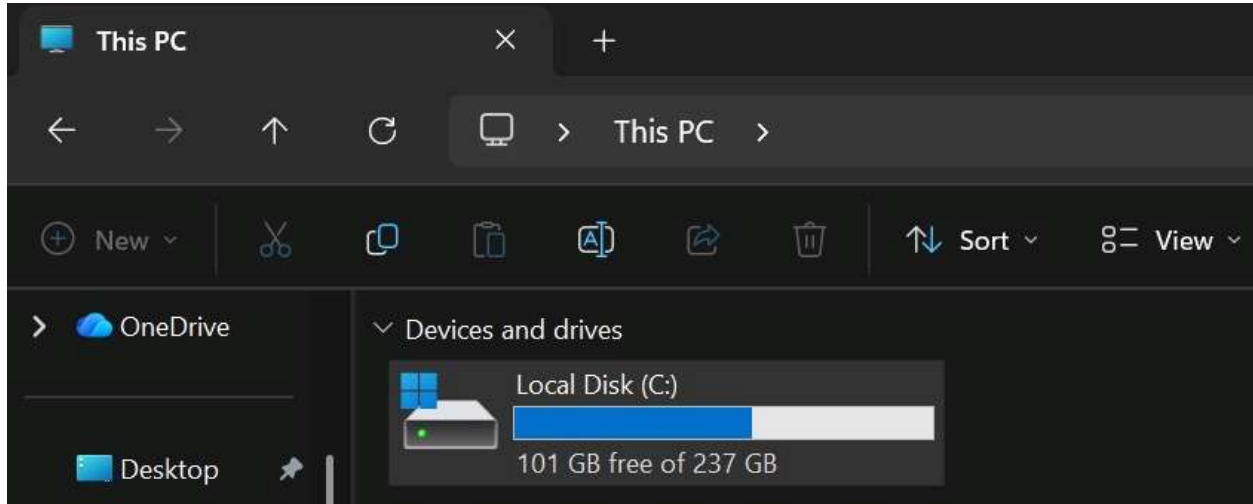
- VMware Workstation Pro installed and running on Windows.



- Downloaded Ubuntu Server ISO image file.



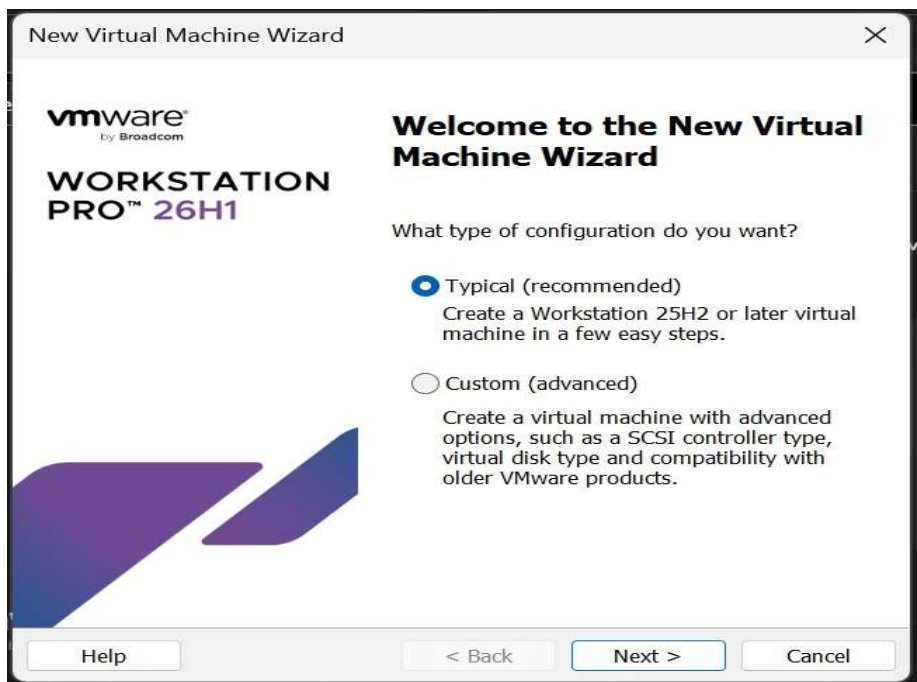
- System storage verification showing >20GB free space.



Task 4:

Create the Ubuntu Server Virtual Machine

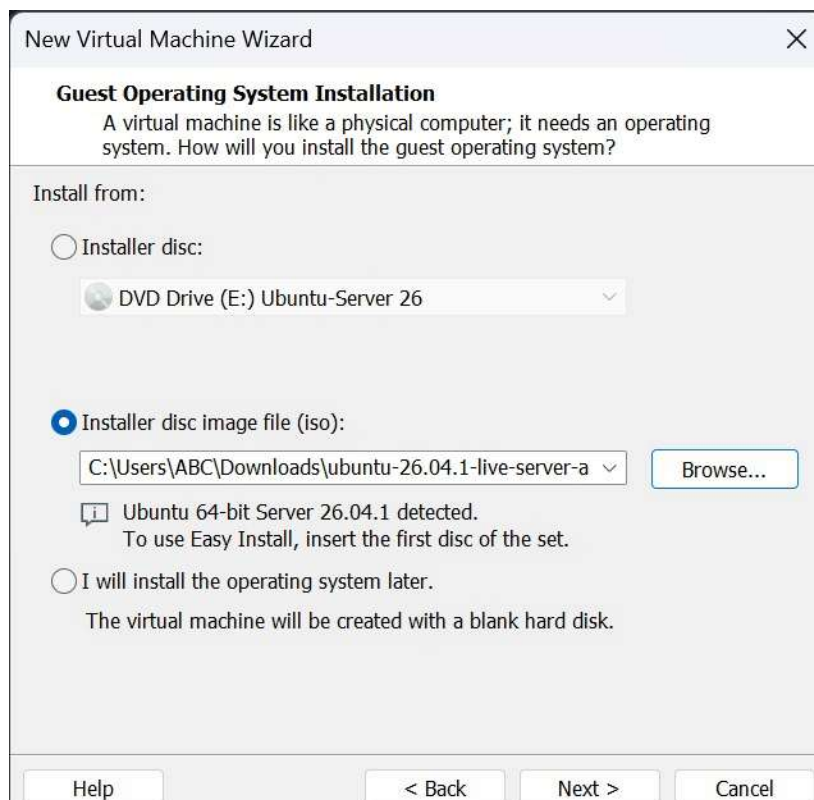
- New Virtual Machine creation wizard in VMware.



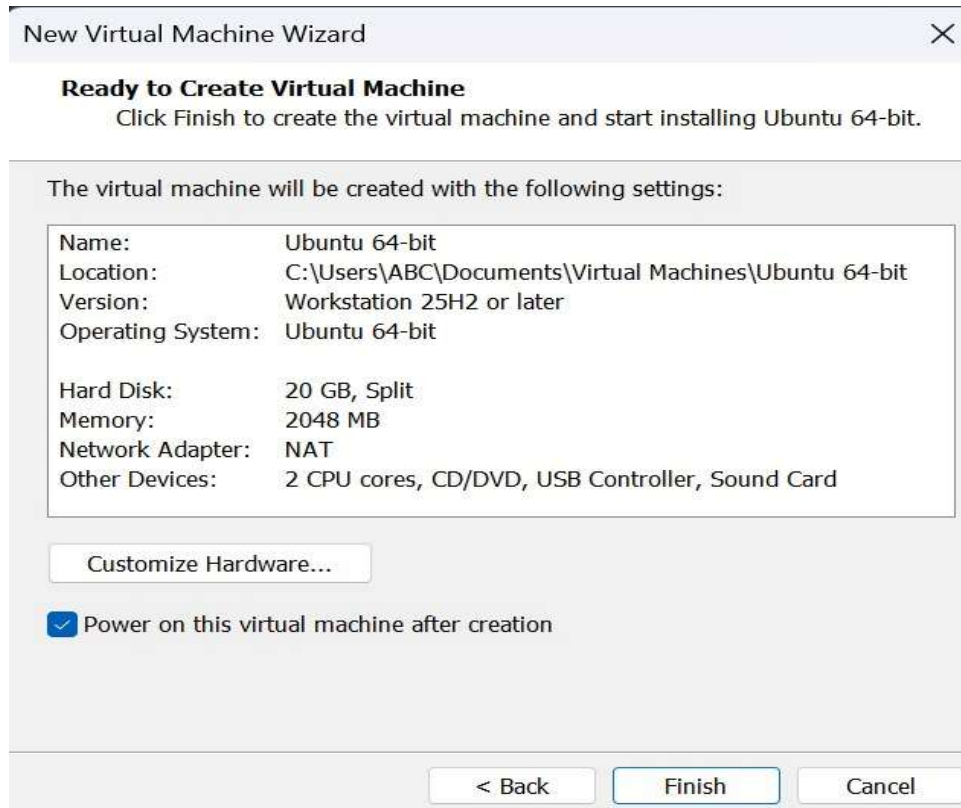
- Selected Typical (recommended) VM configuration.



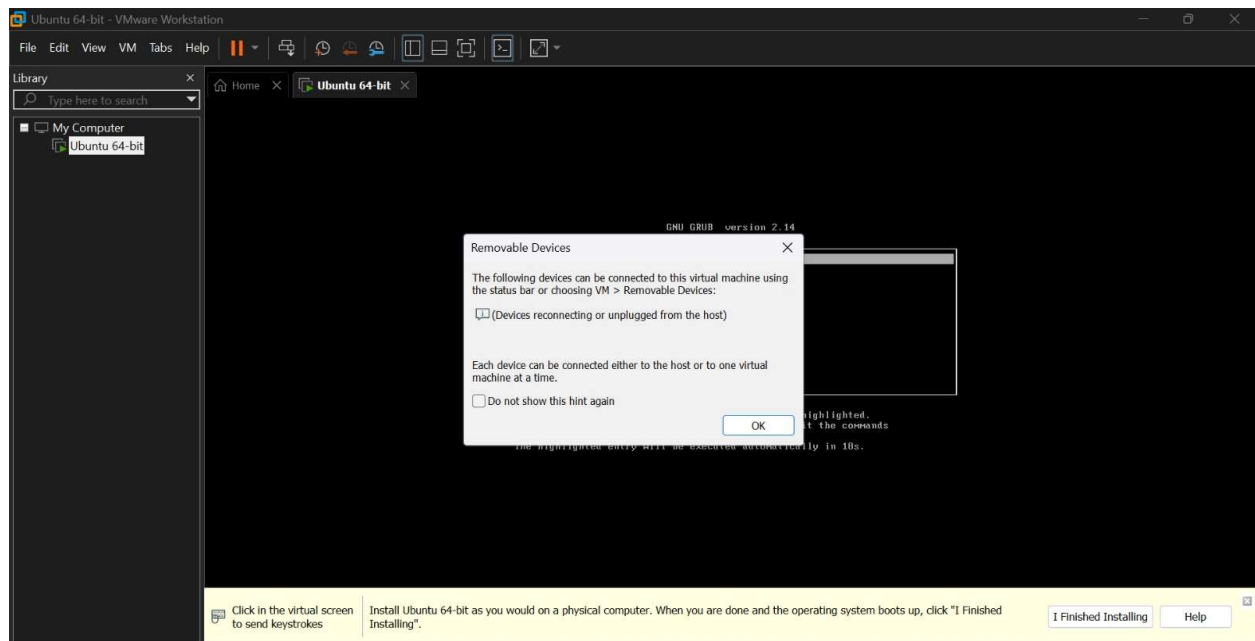
- Ubuntu Server ISO attached to virtual optical drive.



➤ Final Virtual Machine resource summary.



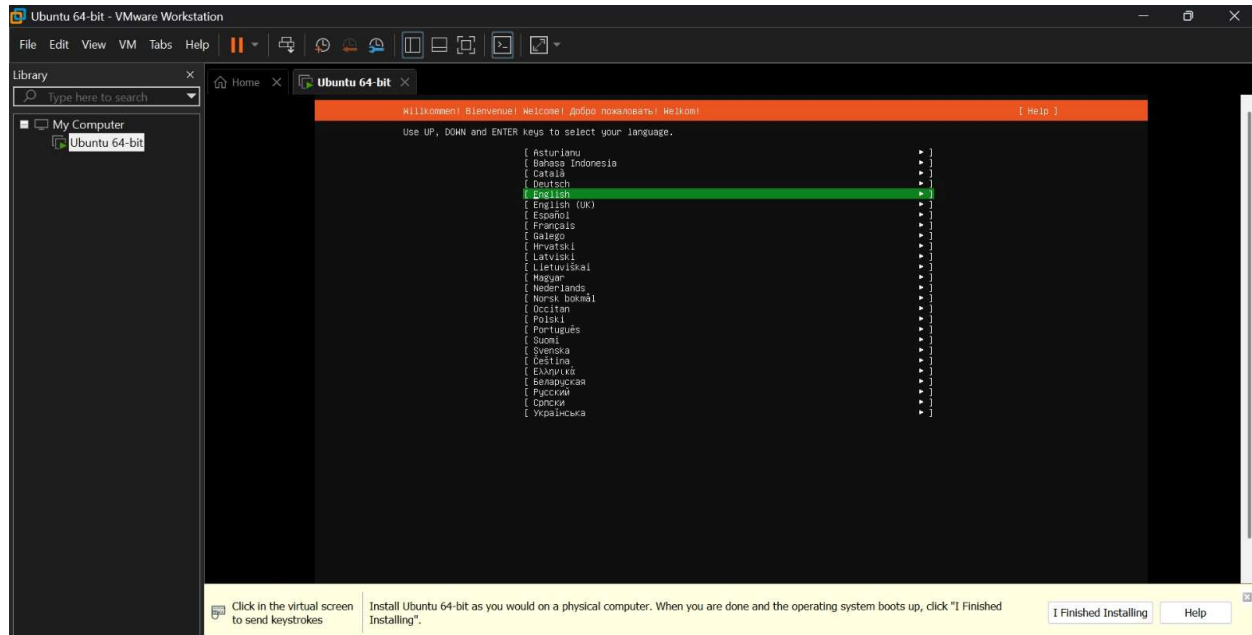
➤ Ubuntu Server initial installer boot screen.



Task 5:

Choose the Installation Language

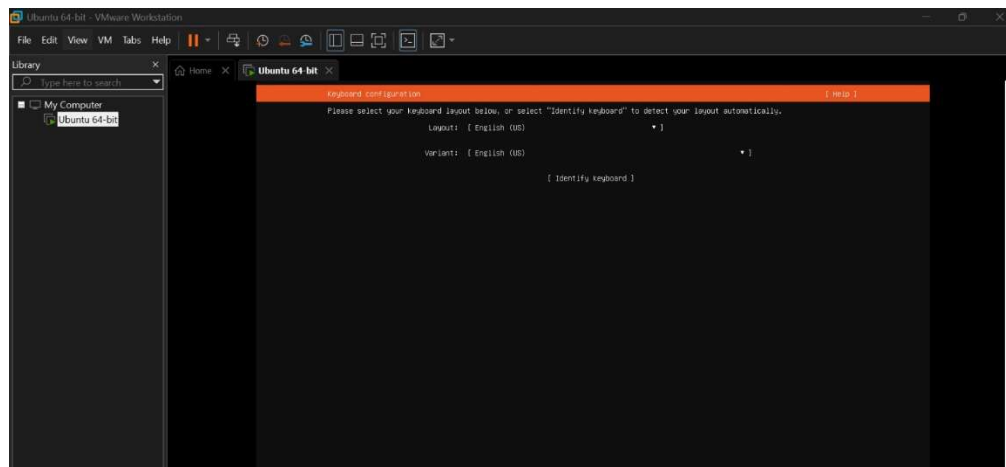
- Selection of preferred installation language (English).



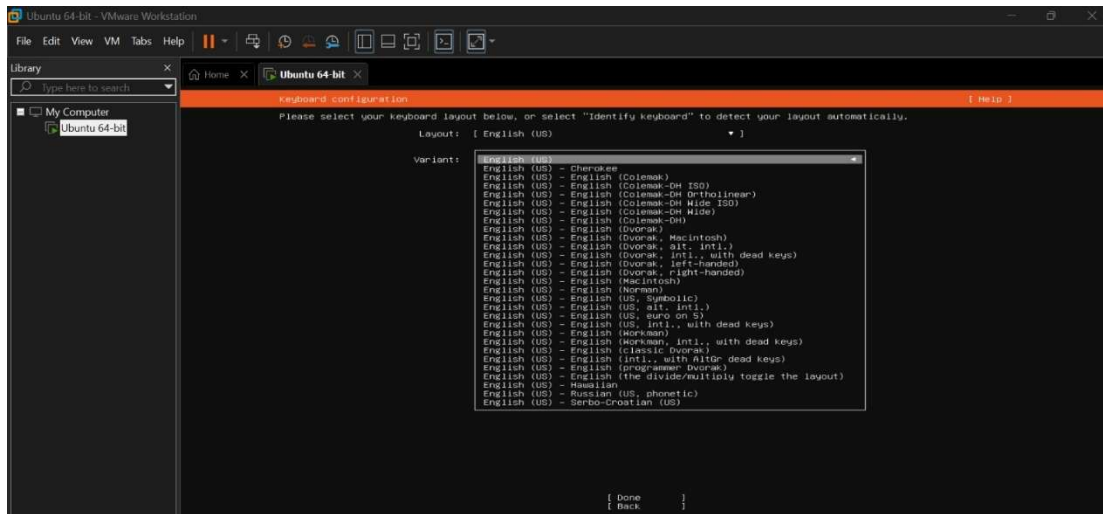
Task 6:

Configure the Keyboard Layout

- Selection of standard keyboard layout.



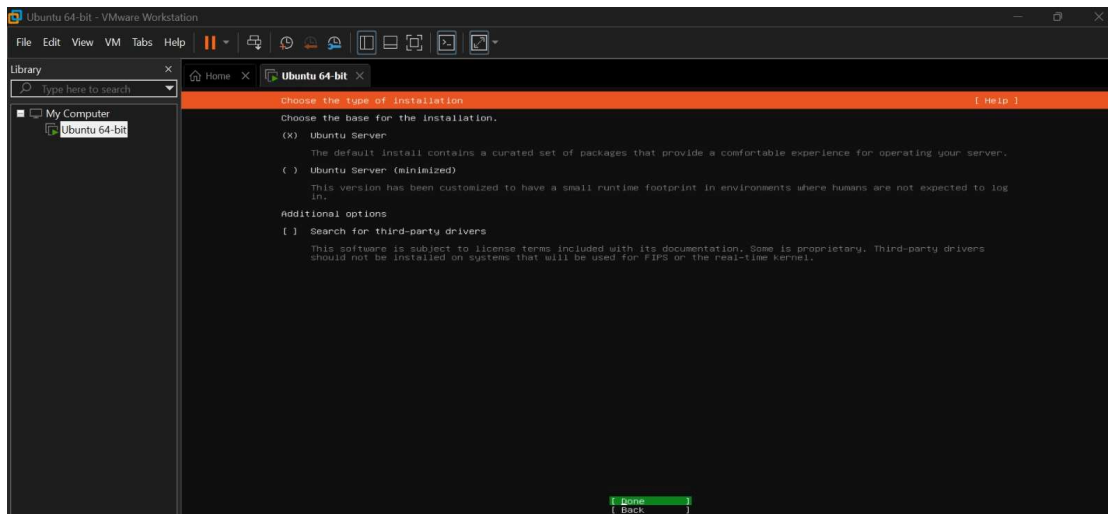
➤ Selection of keyboard variant.



Task 7:

Choose the Ubuntu Server Installation Type

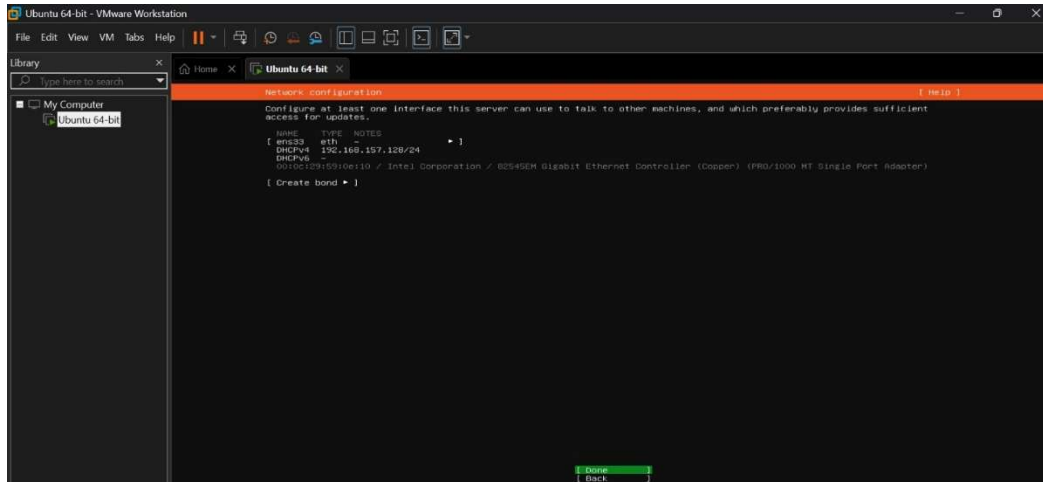
➤ Selected default Ubuntu Server installation base.



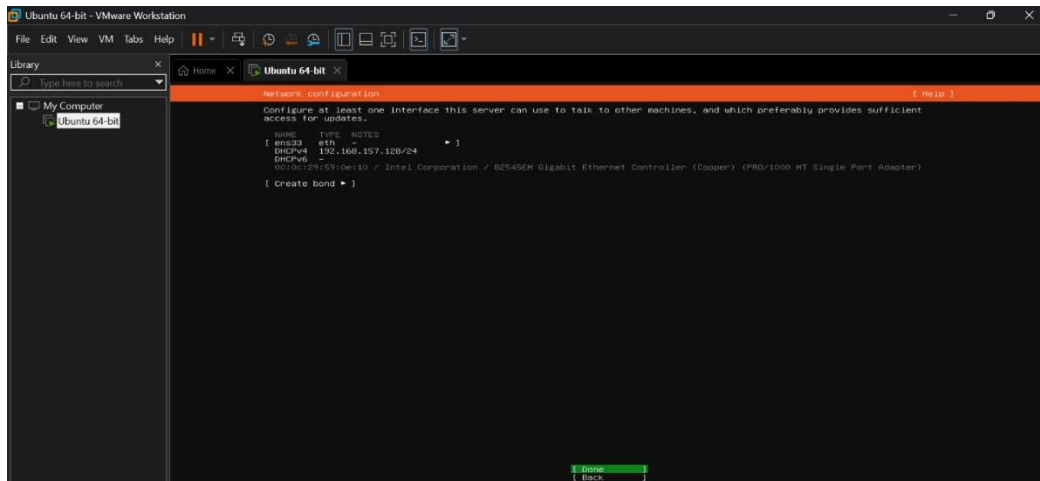
Task 8:

Configure the Network

➤ Caption: Detected active network interface (ens33).



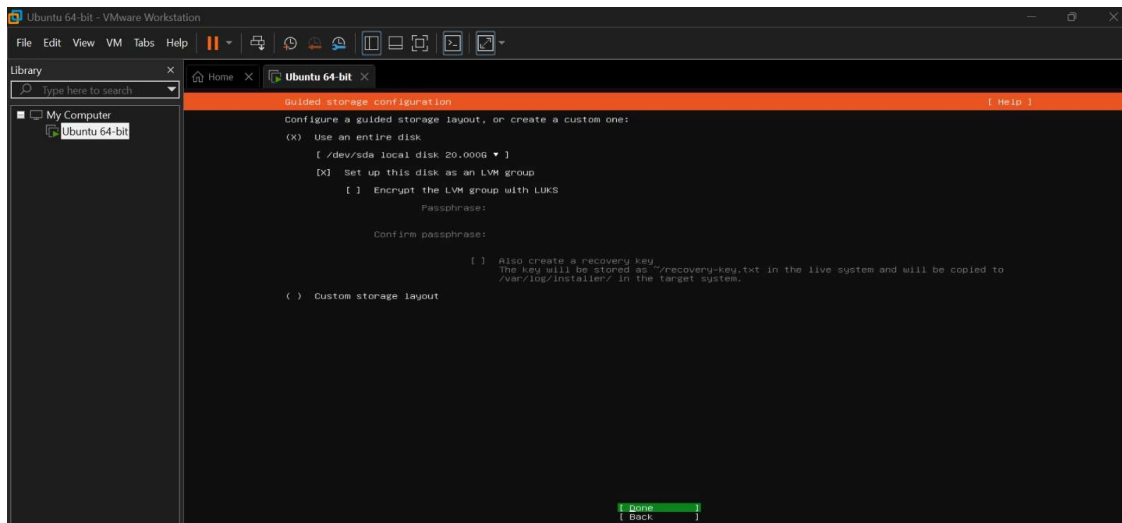
➤ Caption: Dynamic IP address allocation via DHCP.



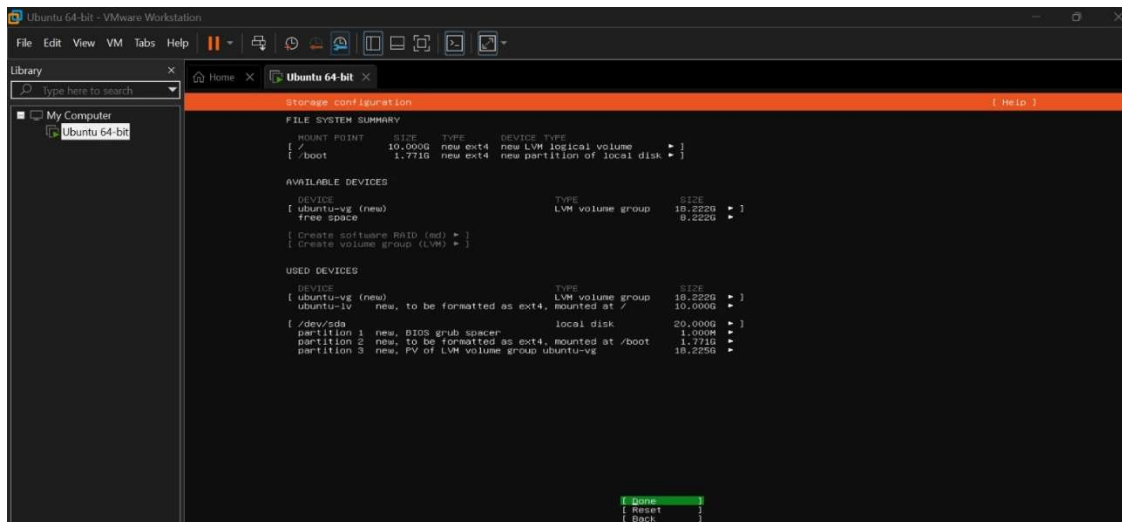
Task 9:

Configure Storage

➤ Selected guided layout on virtual hard drive.



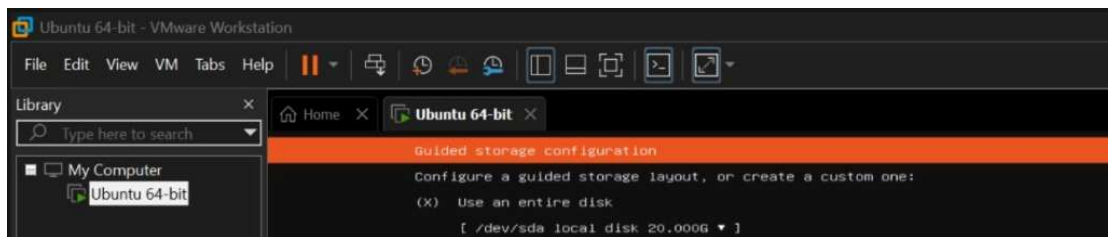
➤ Initial disk partitioning structure review.



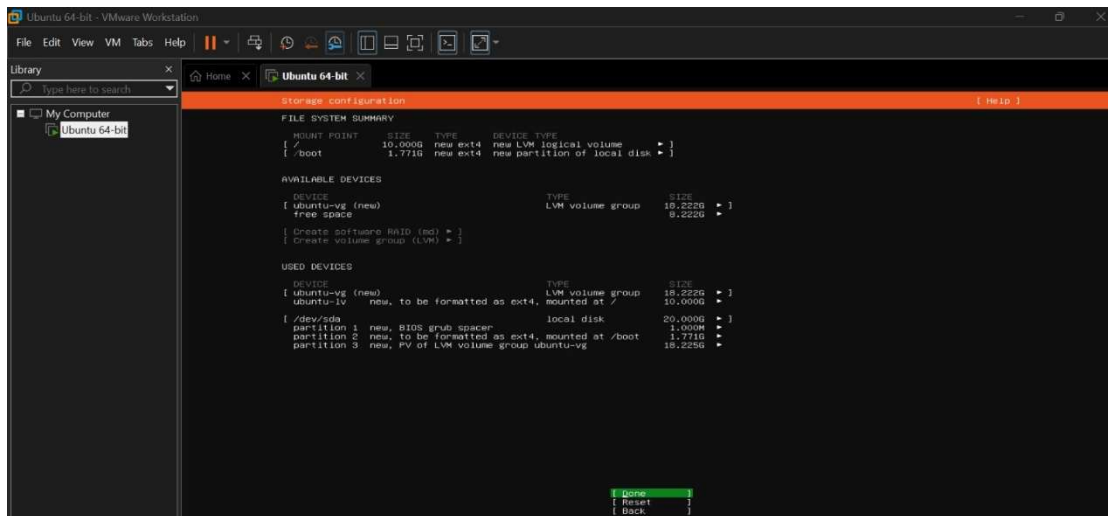
Task 10:

Select the Installation Disk

➤ Selected target VMware virtual disk.



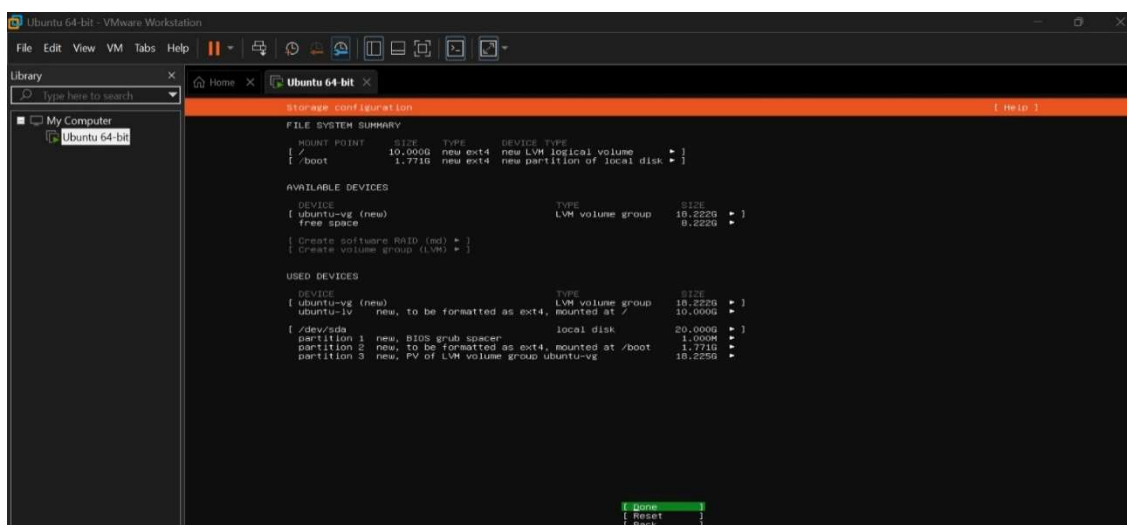
➤ Disk size and target drive verification.



Task 11:

Review the Partitions

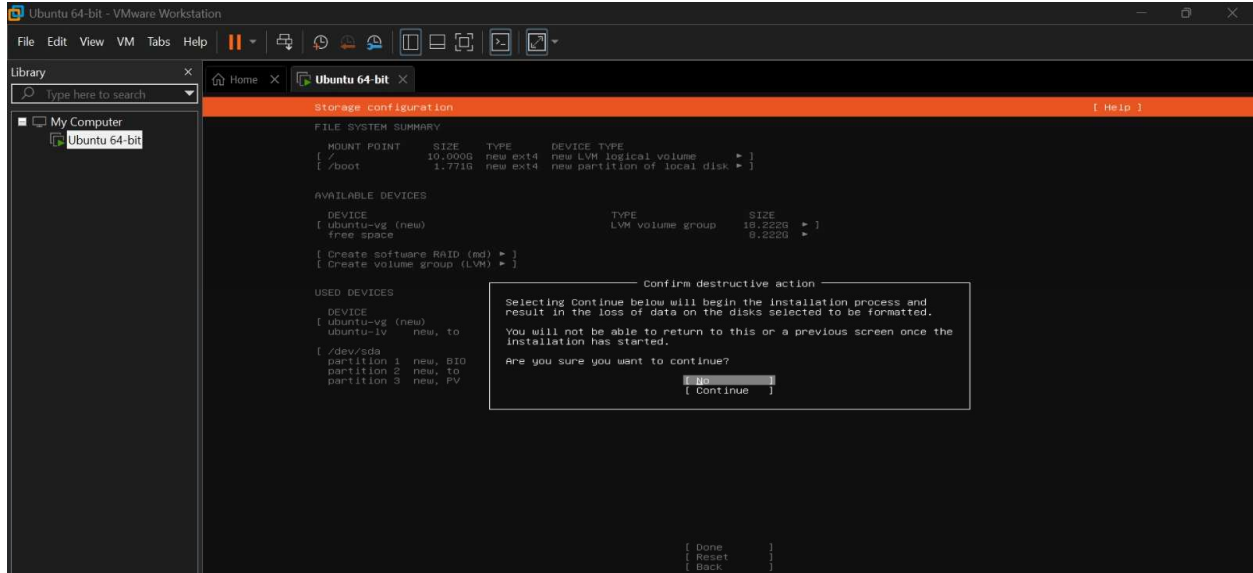
➤ Caption: Final storage partition map before installation.



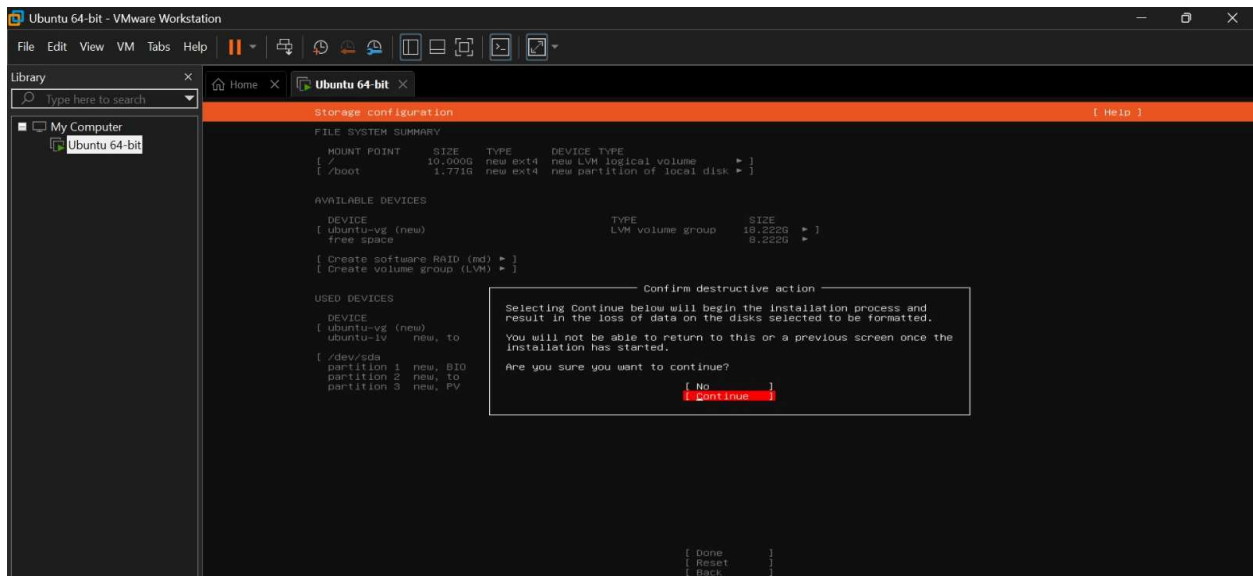
Task 12:

Confirm the Storage Changes

- Installer destructive action warning prompt.



- Formally confirmed disk write execution.



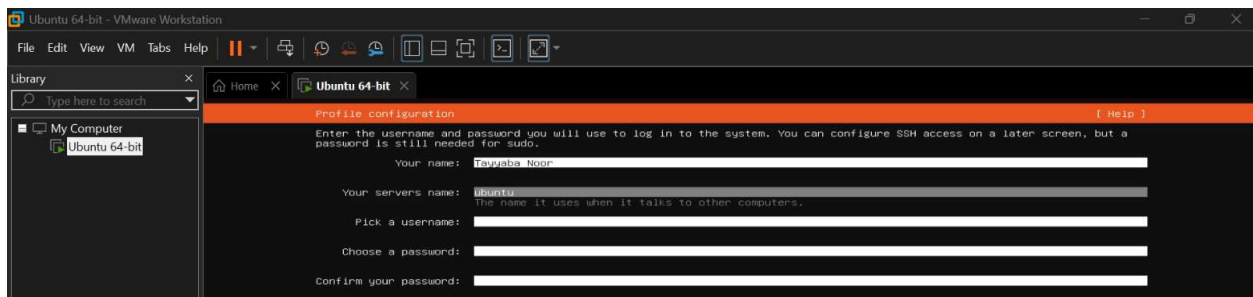
Task 13:

Configure the Ubuntu Profile

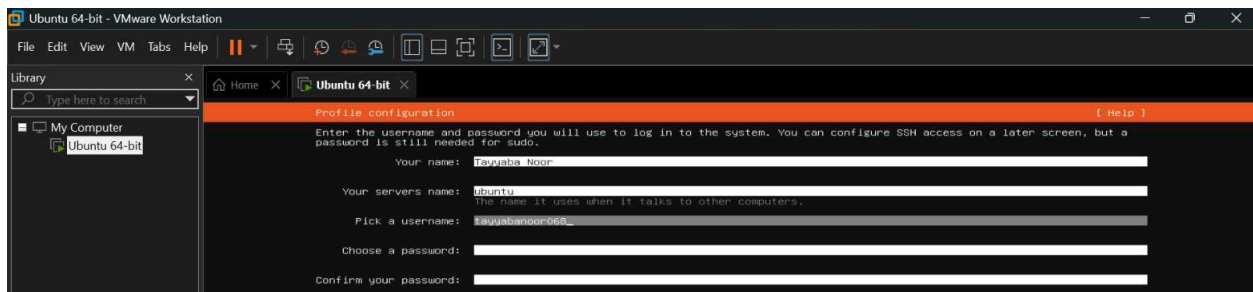
- Entered full actual name in profile setup



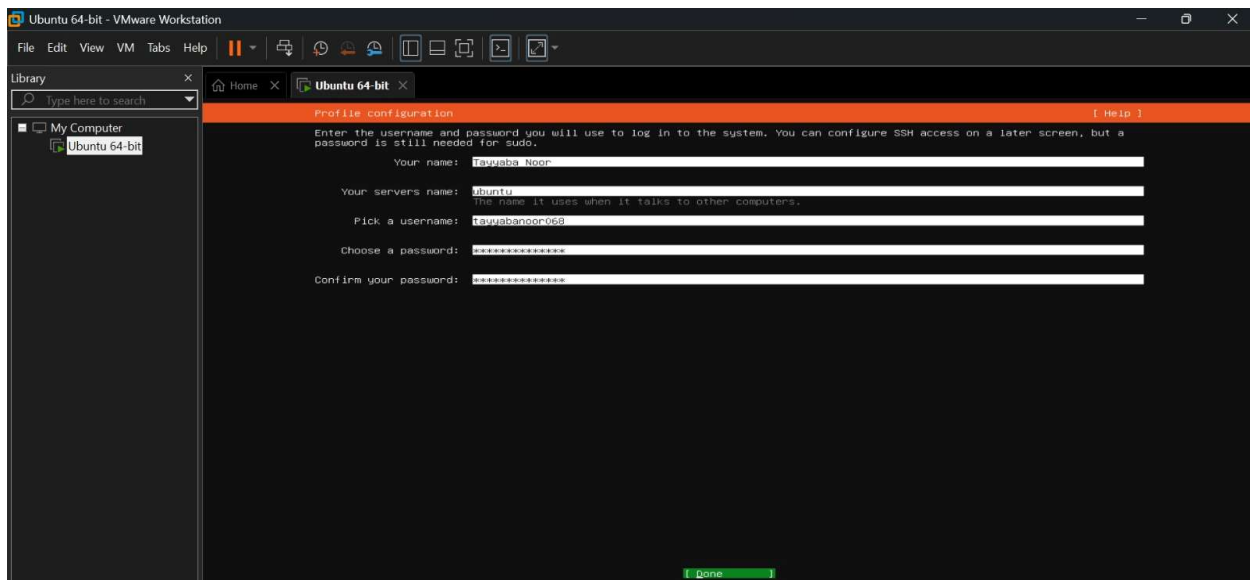
- Set server hostname as ubuntu.



- Set Linux username as tayyabanoor068.



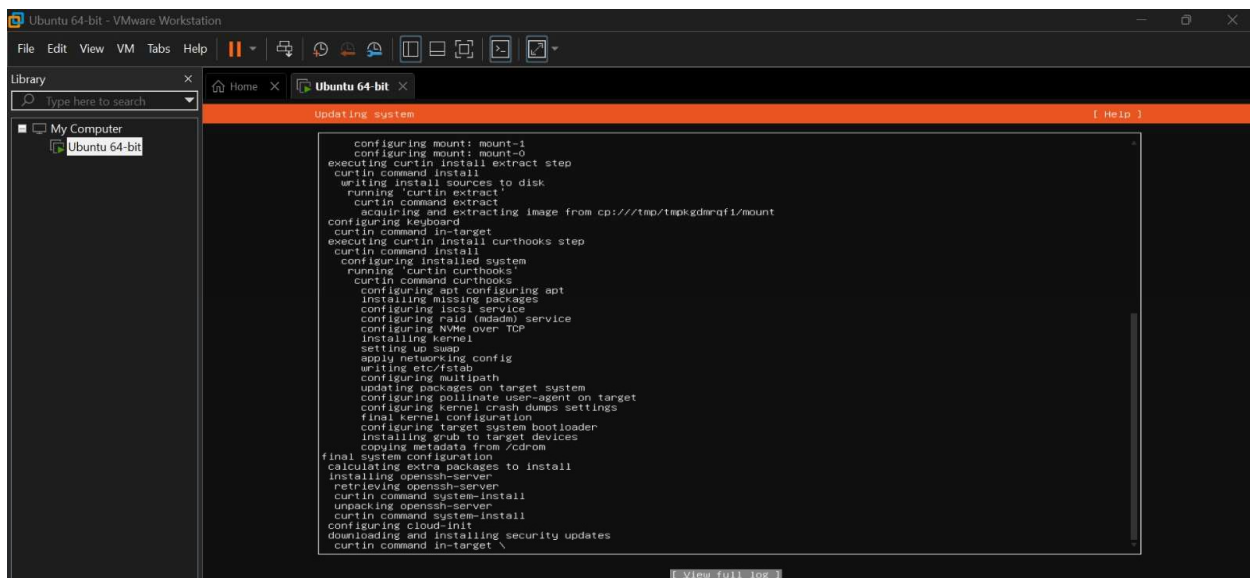
- Complete user credentials summary screen.



Task 14:

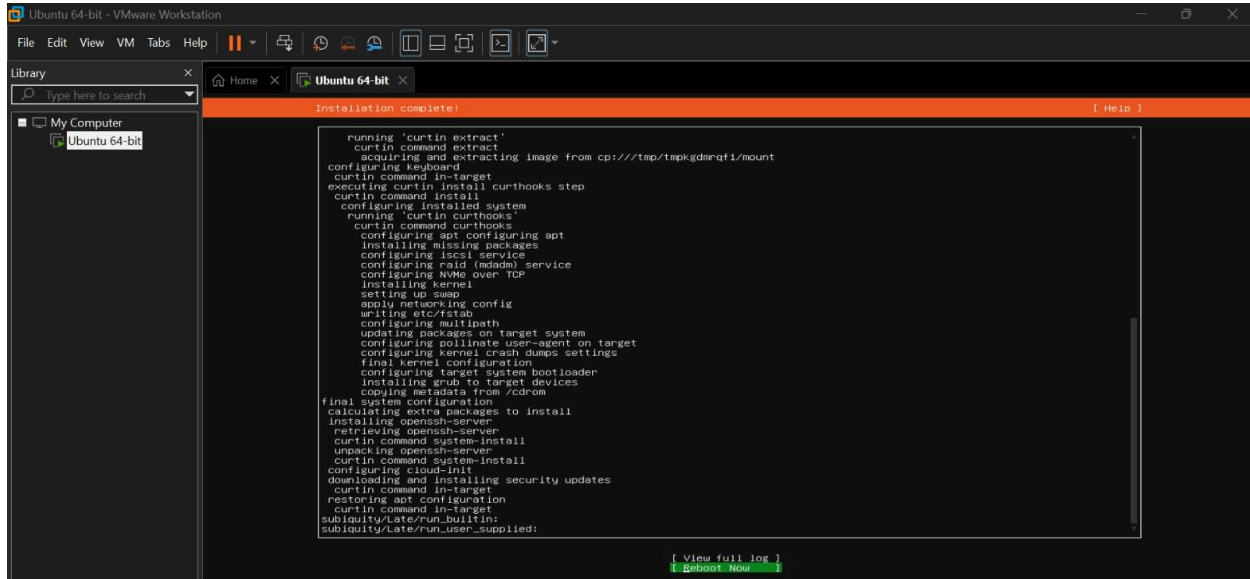
Complete the Software Installation

- Core system packages installation in progress.

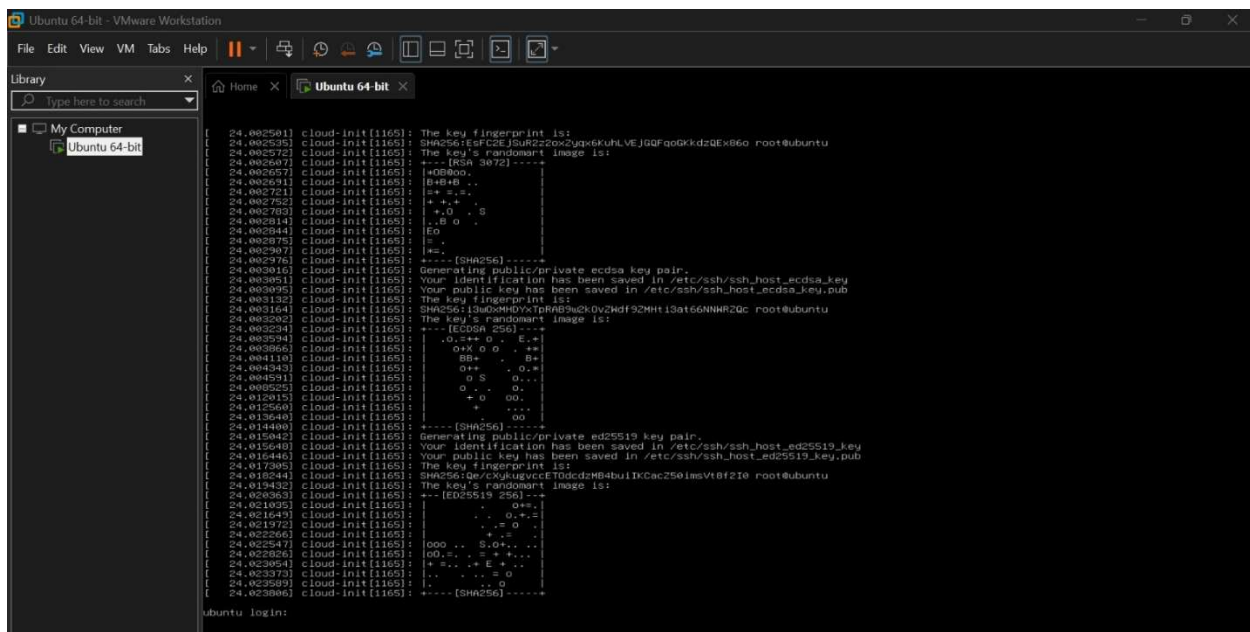


Reboot and Verify the Ubuntu Identity

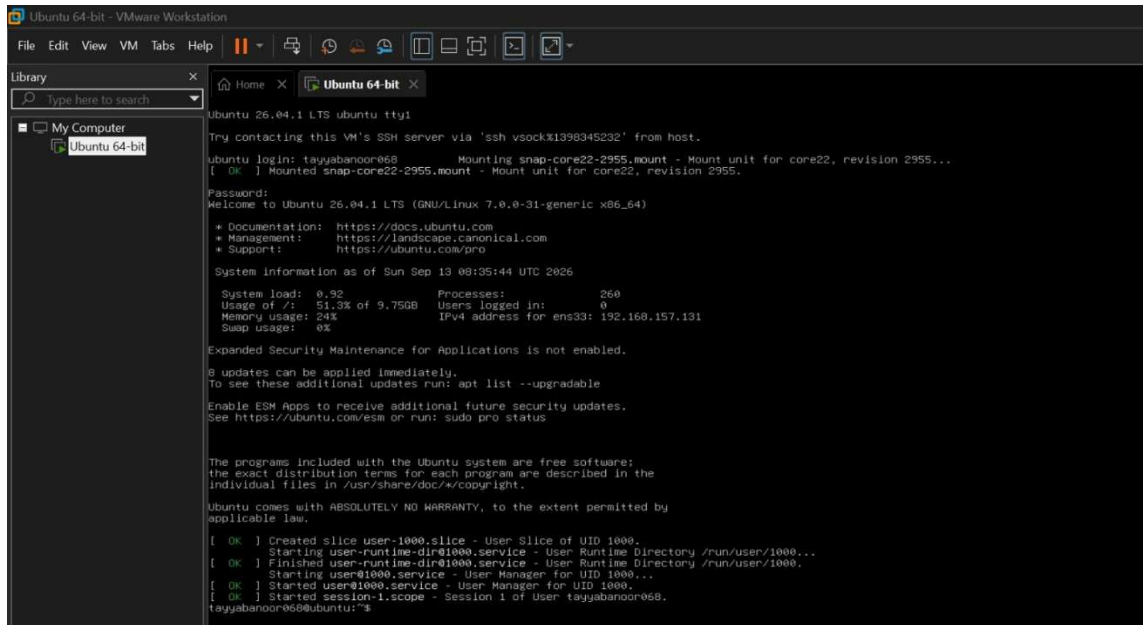
- Installation completed screen showing Reboot option.



- Ubuntu Server login prompt after restart.



➤ Successful system login via local console.



```
Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Ubuntu 64-bit
ubuntu 26.04.1 LTS ubuntu tty1
Try contacting this VM's SSH server via 'ssh vsock:1398345232' from host.
ubuntu login: tayyabanoor068      Mounting snap-core22-2955.mount - Mount unit for core22, revision 2955...
[ OK ] Mounted snap-core22-2955.mount - Mount unit for core22, revision 2955.
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System Information as of Sun Sep 13 08:35:44 UTC 2026

System load:  0.92      Processes:      260
Usage of /:   51.3% of 9.75GB   Users logged in: 0
Memory usage: 24%      IPv4 address for ens33: 192.168.157.131
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

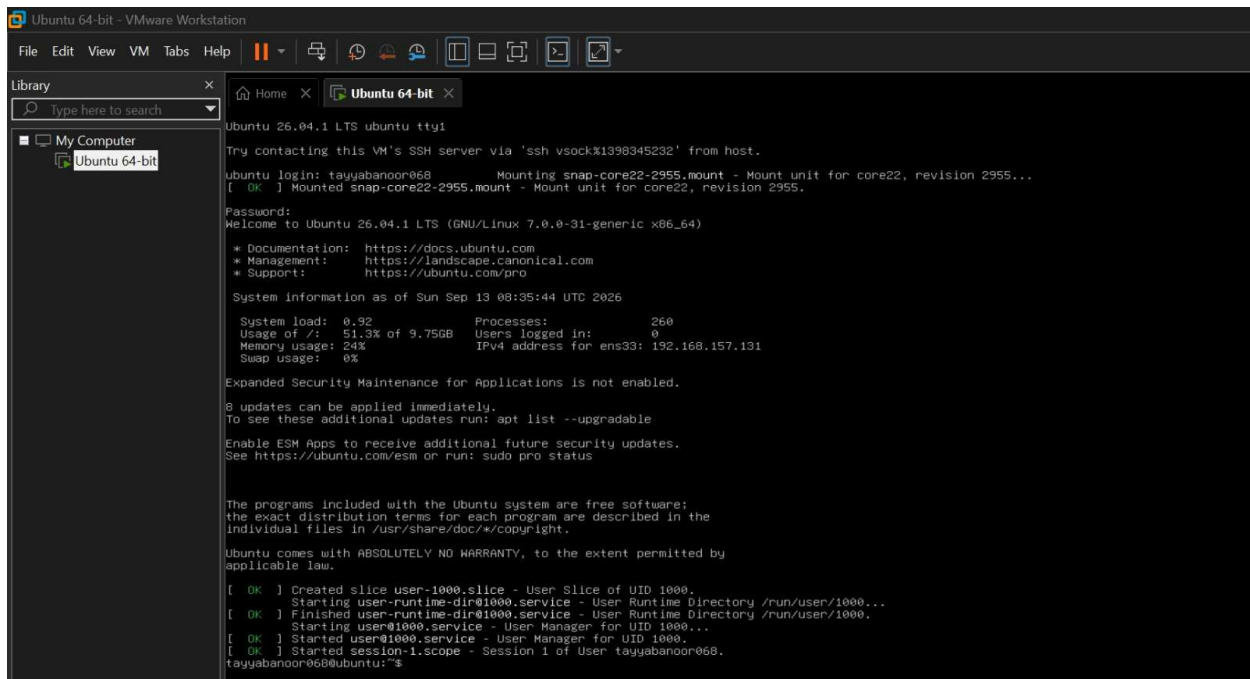
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

[ OK ] Created slice user-1000.slice - User Slice of UID 1000.
Starting user-runtime-dir@1000.service - User Runtime Directory /run/user/1000...
[ OK ] Finished user-runtime-dir@1000.service - User Runtime Directory /run/user/1000.
Starting user@1000.service - User Manager for UID 1000...
[ OK ] Started user@1000.service - User Manager for UID 1000.
[ OK ] Started session-1.scope - Session 1 of User tayyabanoor068.
tayyabanoor068@ubuntu:~$
```

➤ Verified local prompt tayyabanoor068@ubuntu:~\$.



```
Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Ubuntu 64-bit
ubuntu 26.04.1 LTS ubuntu tty1
Try contacting this VM's SSH server via 'ssh vsock:1398345232' from host.
ubuntu login: tayyabanoor068      Mounting snap-core22-2955.mount - Mount unit for core22, revision 2955...
[ OK ] Mounted snap-core22-2955.mount - Mount unit for core22, revision 2955.
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

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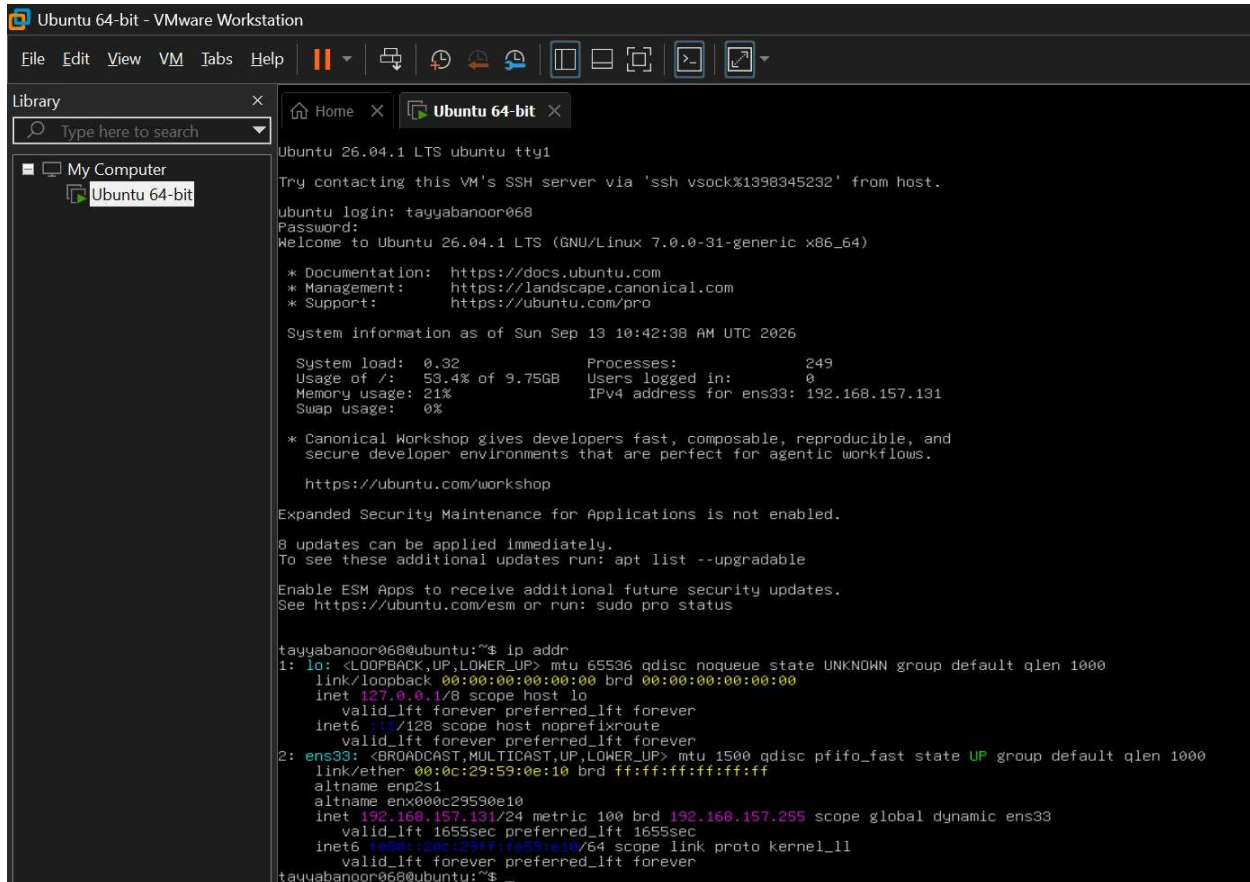
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

[ OK ] Created slice user-1000.slice - User Slice of UID 1000.
Starting user-runtime-dir@1000.service - User Runtime Directory /run/user/1000...
[ OK ] Finished user-runtime-dir@1000.service - User Runtime Directory /run/user/1000.
Starting user@1000.service - User Manager for UID 1000...
[ OK ] Started user@1000.service - User Manager for UID 1000.
[ OK ] Started session-1.scope - Session 1 of User tayyabanoor068.
tayyabanoor068@ubuntu:~$
```

Task 16:

Find the Ubuntu Server IP Address

- Execution of ip addr command in terminal.



```
Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Ubuntu 64-bit
Ubuntu 26.04.1 LTS ubuntu tty1
Try contacting this VM's SSH server via 'ssh vsock%1398345232' from host.
ubuntu login: tayyabanoor068
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sun Sep 13 10:42:38 AM UTC 2026

System load:  0.32               Processes:    249
Usage of /:   53.4% of 9.75GB    Users logged in: 0
Memory usage: 21%               IPv4 address for ens33: 192.168.157.131
Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.
   https://ubuntu.com/workshop

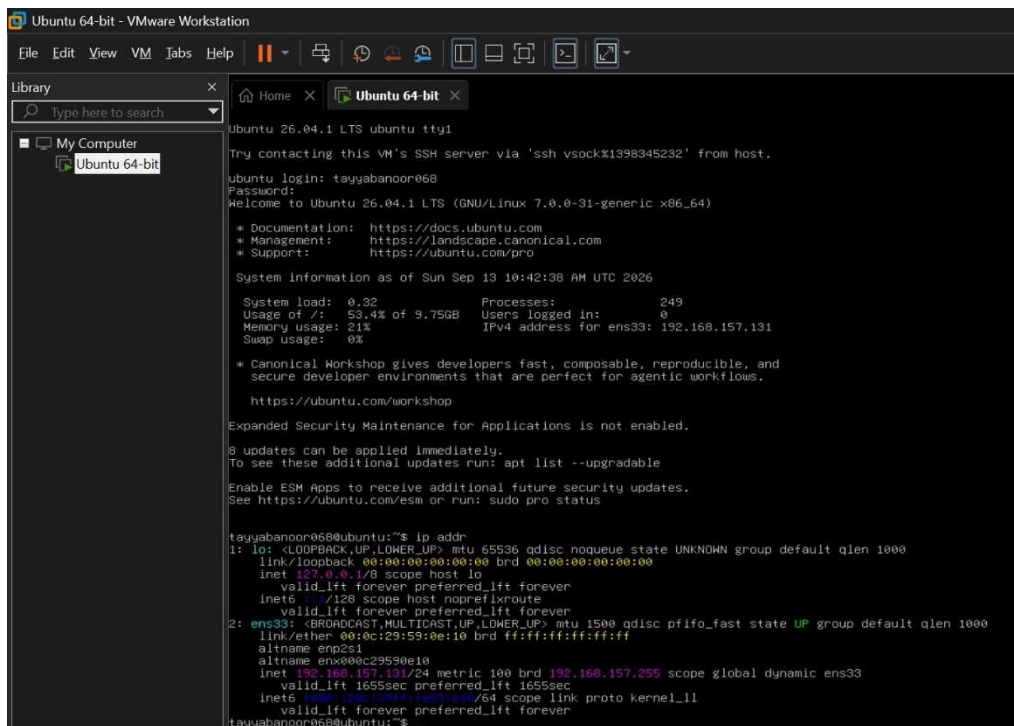
Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

tayyabanoor068@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:59:0e:10 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altnamex enx000c29590e10
    inet 192.168.157.131/24 metric 100 brd 192.168.157.255 scope global dynamic ens33
        valid_lft 1655sec preferred_lft 1655sec
    inet6 fe80::20c:29:59:10/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
tayyabanoor068@ubuntu:~$
```

- Identified IPv4 address (192.168.157.131) under ens33.



```
Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Ubuntu 64-bit
ubuntu 26.04.1 LTS ubuntu tty1
Try contacting this VM's SSH server via 'ssh vsockx1398345232' from host.
ubuntu login: tayyabanoor068
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
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System information as of Sun Sep 13 10:42:38 AM UTC 2026

System load:  0.32          Processes:      249
Usage of /:   53.4% of 9.75GB Users logged in:    0
Memory usage: 21%          IPv4 address for ens33: 192.168.157.131
Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.
   https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
To see these additional updates run: apt list --upgradable
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

tayyabanoor068@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:59:0e:10 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c29590e10
    inet 192.168.157.131/24 metric 100 brd 192.168.157.255 scope global dynamic ens33
        valid_lft 1655sec preferred_lft 1655sec
    inet6 ::2/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
tayyabanoor068@ubuntu:~$
```

Task 17:

Connect to Ubuntu Server from Windows

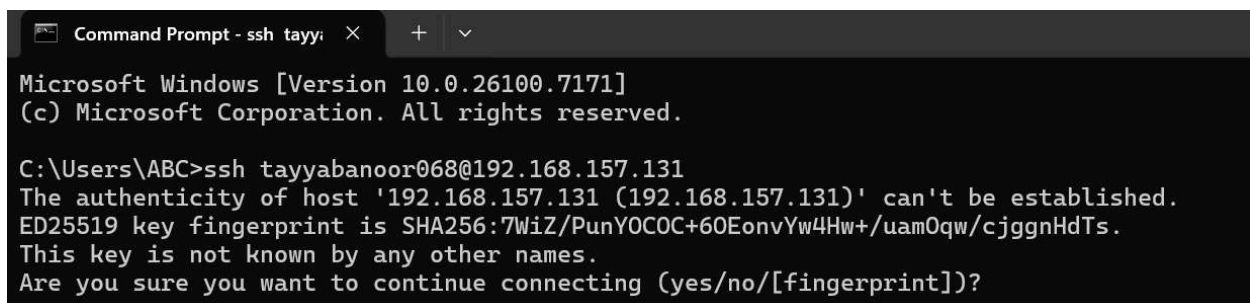
- SSH connection command initiated from Windows CMD.



```
Command Prompt
Microsoft Windows [Version 10.0.26100.7171]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ABC>ssh tayyabanoor068@192.168.157.131
```

- Host key fingerprint authorization prompt (yes).



```
Command Prompt - ssh tayy:
Microsoft Windows [Version 10.0.26100.7171]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ABC>ssh tayyabanoor068@192.168.157.131
The authenticity of host '192.168.157.131 (192.168.157.131)' can't be established.
ED25519 key fingerprint is SHA256:7WiZ/PunYOCOC+60EonvYw4Hw+/uam0qw/cjggnHdTs.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
```

➤ Successful SSH login into Ubuntu Server from Windows.

```
tayyabanoor068@ubuntu: ~  
Warning: Permanently added '192.168.157.131' (ED25519) to the list of known hosts.  
ssh_dispatch_run_fatal: Connection to 192.168.157.131 port 22: Connection aborted  
  
C:\Users\ABC>ssh tayyabanoor068@192.168.157.131  
tayyabanoor068@192.168.157.131's password:  
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Sun Sep 13 10:59:41 AM UTC 2026  
  
System load:  0.0      Processes:            223  
Usage of /:   53.9% of 9.75GB   Users logged in:     0  
Memory usage: 23%      IPv4 address for ens33: 192.168.157.131  
Swap usage:   0%  
  
* Canonical Workshop gives developers fast, composable, reproducible, and  
  secure developer environments that are perfect for agentic workflows.  
  
https://ubuntu.com/workshop  
  
Expanded Security Maintenance for Applications is not enabled.  
  
8 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
tayyabanoor068@ubuntu:~$
```

➤ Remote terminal prompt active as tayyabanoor068@ubuntu:~\$.

```
tayyabanoor068@ubuntu: ~  
Warning: Permanently added '192.168.157.131' (ED25519) to the list of known hosts.  
ssh_dispatch_run_fatal: Connection to 192.168.157.131 port 22: Connection aborted  
  
C:\Users\ABC>ssh tayyabanoor068@192.168.157.131  
tayyabanoor068@192.168.157.131's password:  
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)  
  
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Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
tayyabanoor068@ubuntu:~$
```

Task 18:

➤ Prepare Lab1 Solution Pdf



Lab #01

Github Identity, Student Portal Registration, and Ubuntu Server Installation

Name: Tayyaba Noor

Registration No: FA24B1-SE-068

Course Title: Cloud Computing

Course Code: BSE-410

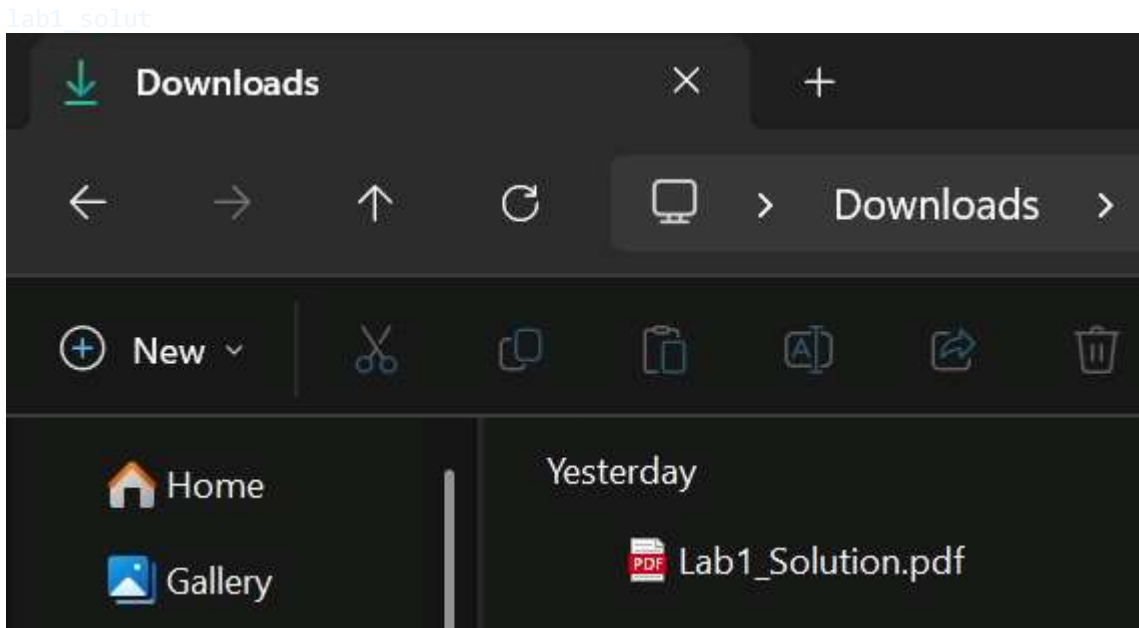
Section: B

Session: 2024-2028

Github Username: tayyabanoor068

Fatima Jinnah Women University
Rawalpindi

➤ Exported Lab1 Solution pdf.



Screenshot demonstrating the exported Lab1_Solution.pdf file saved on the local system.

[illegible]